SmartVision: Traffic Flow Monitoring with YOLO

INTRODUCTION

- This project aims to address these challenges by leveraging computer vision techniques—specifically, **YOLO (You Only Look Once)** object detection—to develop an intelligent and automated system for **real-time road traffic monitoring** and **automobile counting**. YOLO is a deep learning model known for its speed and accuracy in detecting multiple objects in a single frame, making it highly suitable for dynamic traffic environments.

OBJECTIVE

- Developing and implementing a YOLO-based object identification model that can locate and identify traffic-related items in urban driving situations is the main goal of this research.

- This yolo was designed to process real-time input images and accurately detect multiple classes of objects such as vehicles, pedestrians, and traffic signals using a streamlined, efficient, and lightweight architecture.

YOLO

1. YOLO (You Only Look Once): What is it?
   YOLO is one of the most advanced real-time object detection algorithms.
2. YOLO is incredibly quick and effective since it processes photos in a single neural network pass rather than in several steps like other systems do (such as region recommendations followed by categorization).

   Important attributes:

   Identifies items and their locations simultaneously.

   Divides an image into a grid and predicts bounding boxes and class probabilities directly.

   Results are output in the [class_id] format. [x_center] [y_center] [height] [width]
   Normalized values range from 0 to 1.

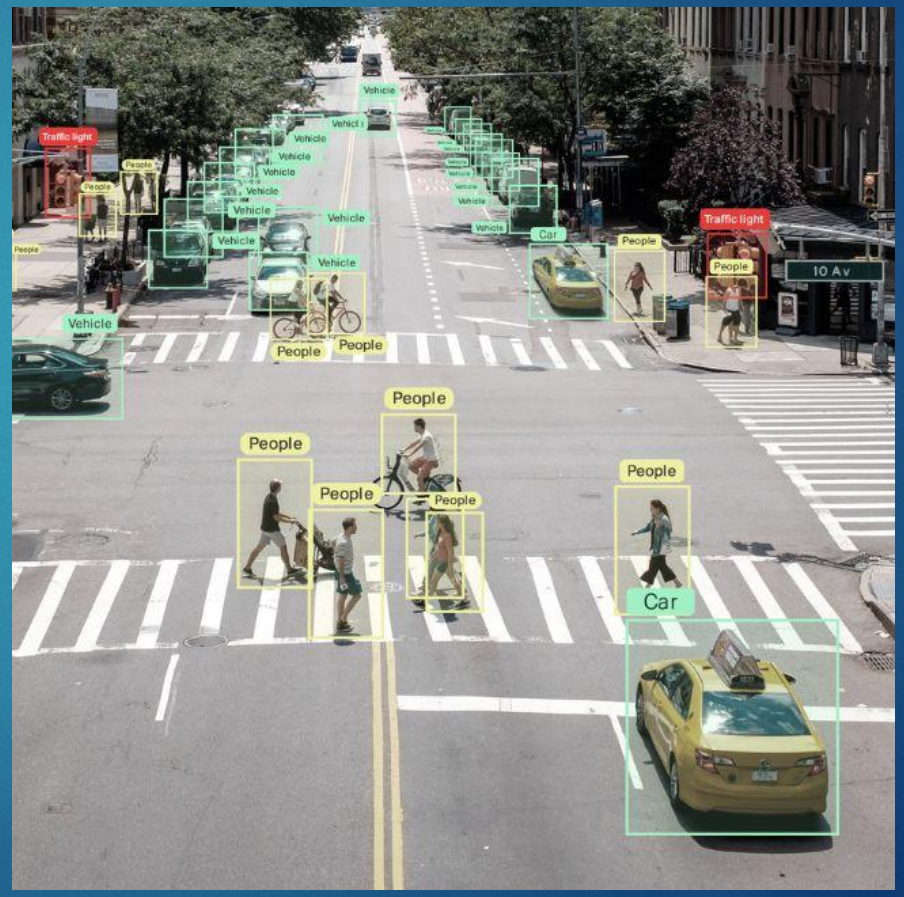
YOLO APPLICATIONS

Because of its quickness and reasonable accuracy, YOLO is often used in embedded and real-time applications.
Typical usage cases include of:

- Autonomous Vehicles: Recognizing traffic signals, road signs, pedestr ians, and other vehicles.
- Surveillance Systems: Tracking people and things in real time.
- Industrial automation: identifying items or flaws in production lines.
- Robotics: Identifying objects for manipulation and navigation.
- Retail analytics: tracking the movements and actions of customers.
- Augmented Reality: AR technologies that overlay objects in real tim e.

EXAMPLE:

Surveillance Systems



How YOLO Works:

1. Divide Image into Grid: Image is split into S x S grid cells.
2. Each Grid Cell Predicts:
   - Bounding box coordinates.
   - Confidence score (likelihood an object exists in the box).
   - Class probabilities.
3. Prediction Output: All predictions are made in one forward pass.
4. Post-processing:Apply Non-Maximum Suppression (NMS) to remove overlapping boxes.

ADVANTAGES OF YOLO

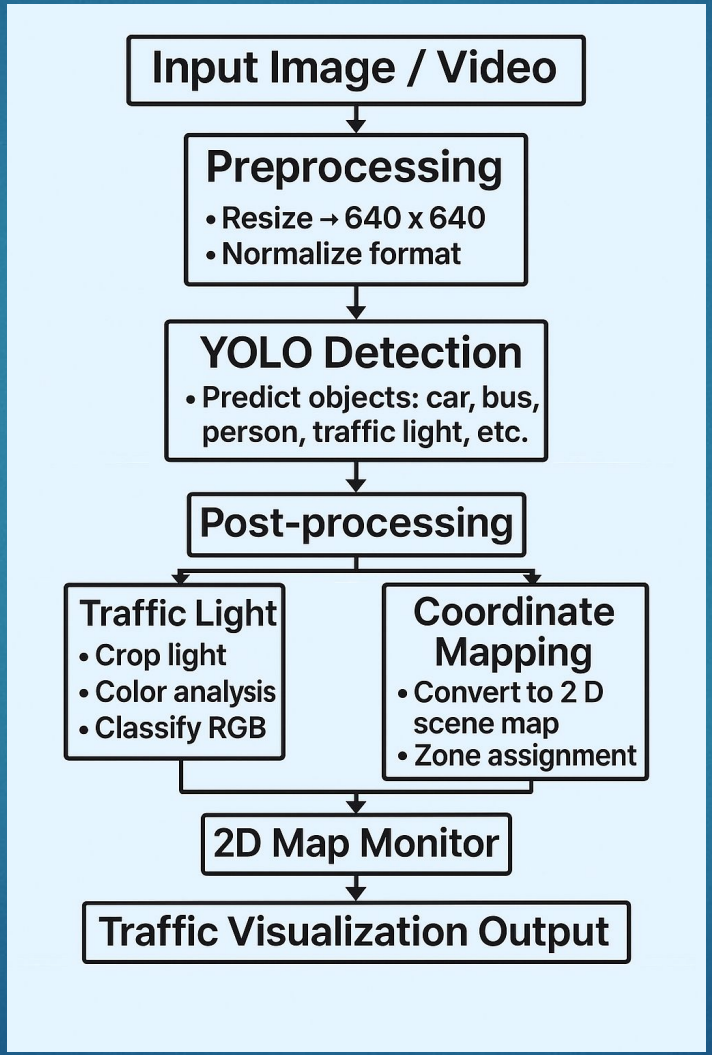
- Real-time Performance: Extremely fast — suitable for video and real-time systems.
- Single-Step Detection: Unified architecture simplifies the pipeline.
- Good Accuracy: Especially in newer versions (YOLOv7 to YOLOv11).
- Lightweight Versions Available: (like YOLOv10 Nano) for mobile or embedded devices.
- Open-source Ecosystem: Large community and many pre-trained models available.

Architecture

- Input Image: Sourced from online video data or live stream.
- Preprocessing: Images resized to 640x640, formatted for YOLO input.
- YOLO Inference: Model returns class IDs and bounding boxes for 10 objects.
- Traffic Light Analysis .
- Bounding box cropped- .
- Dominant color extracted using OpenCV.
- Classified into red/yellow/green states.
- 2D Mapping & Zone Tracking.
- Bounding boxes projected into scene layout.
- Mapped to grid or minimap coordinates.
- Helps track movement and spatial distribution.

- Post-YOLO Modules and Purpose.
- Traffic Light Detection.
  - Uses bounding box region → detects current light status.
  - Helps enforce traffic logic (e.g., stop detection).
  - Enables automatic rule violation monitoring.
- 2D Mapping System.
  - Converts pixel positions → real-world layout zones.
  - Helps visualize where objects are, detect congestion zones.
  - Useful for understanding flow, occupancy, and behavior.

ARCHITECTURE FLOWCHART



COMPARATIVE ANALYSIS

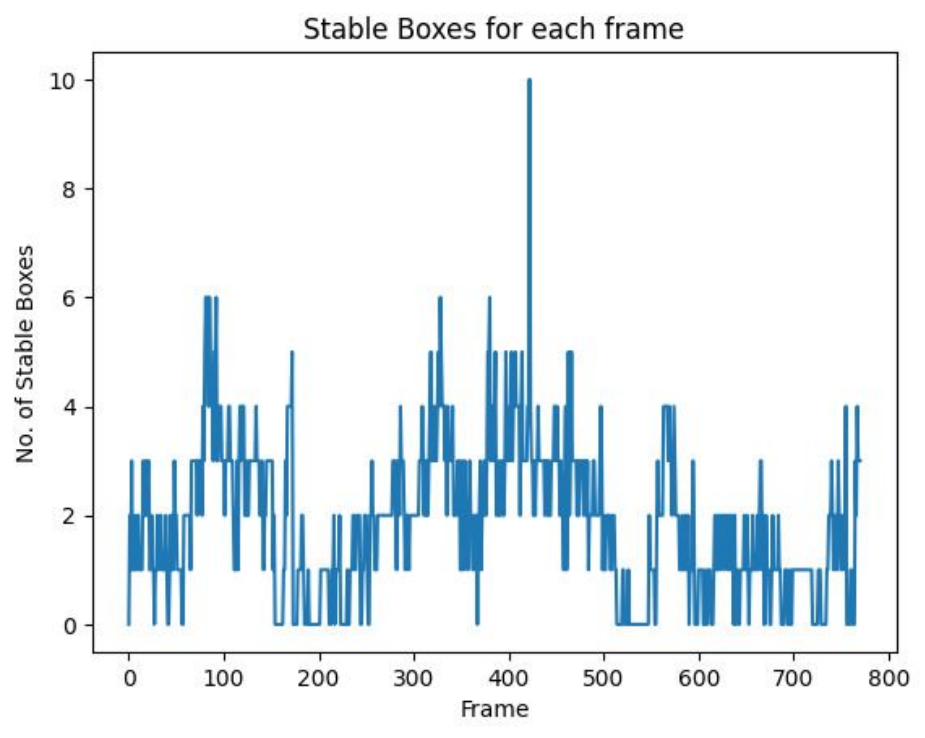
Yolo versions v8, v9 , v10 and v11 Nano versions were compared for our application for the following parameters :

- Accuracy of Object Detection with respect to manual observation
- Bounding Box Consistency
- Average Stable Boxes per Frame

Parameters

ACCURACY (Measured Manually)

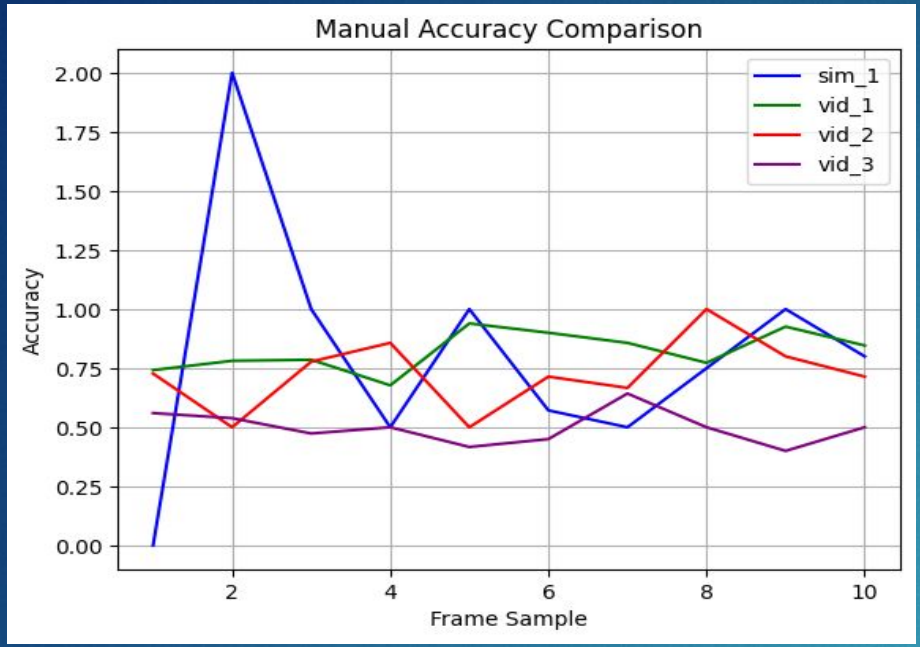
- Random Sampling of Frames
- Calculated for 4 videos for each yolo version
- Factors affecting this parameter
  - Frames per second
  - Stable boxes in Frames



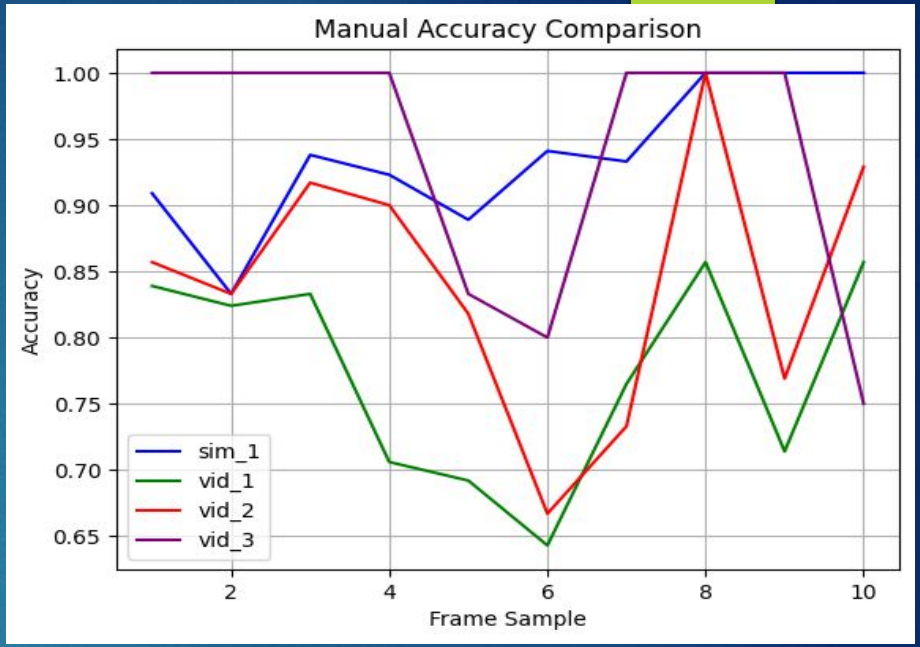
Calculation

- For each frame ,
- (no. of detections by yolo) / (no. of manual identifications)

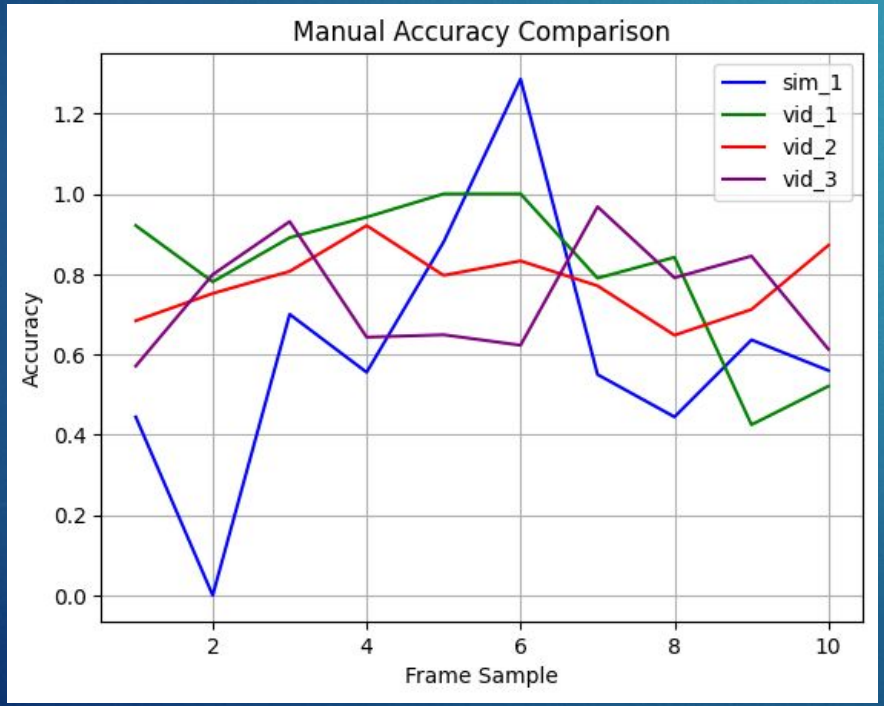
V8


Manual Accuracy Comparison
sim_1
vid_1
vid_2
vid_3
Accuracy
Frame Sample

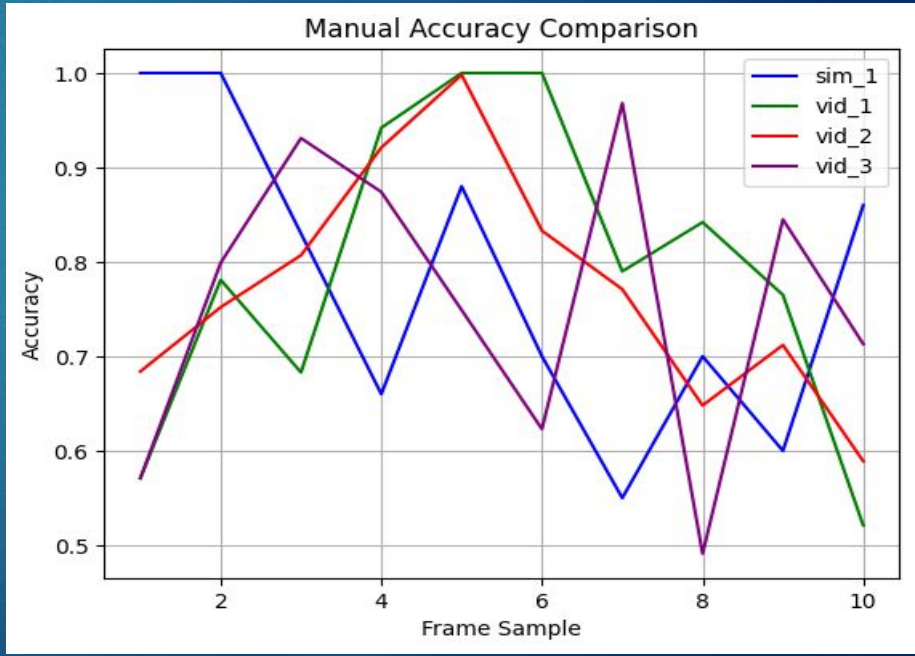
V9


Manual Accuracy Comparison
sim_1
vid_1
vid_2
vid_3
Accuracy
Frame Sample

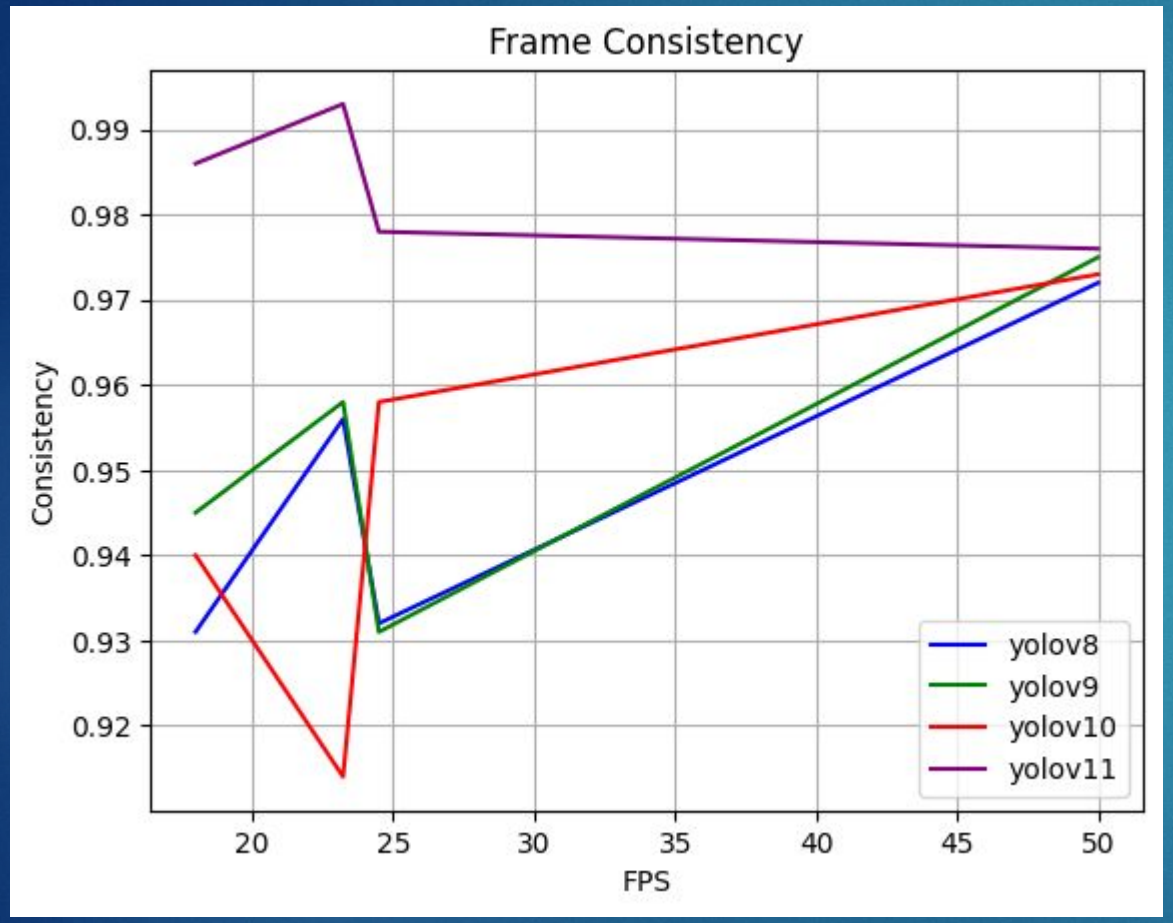
V10

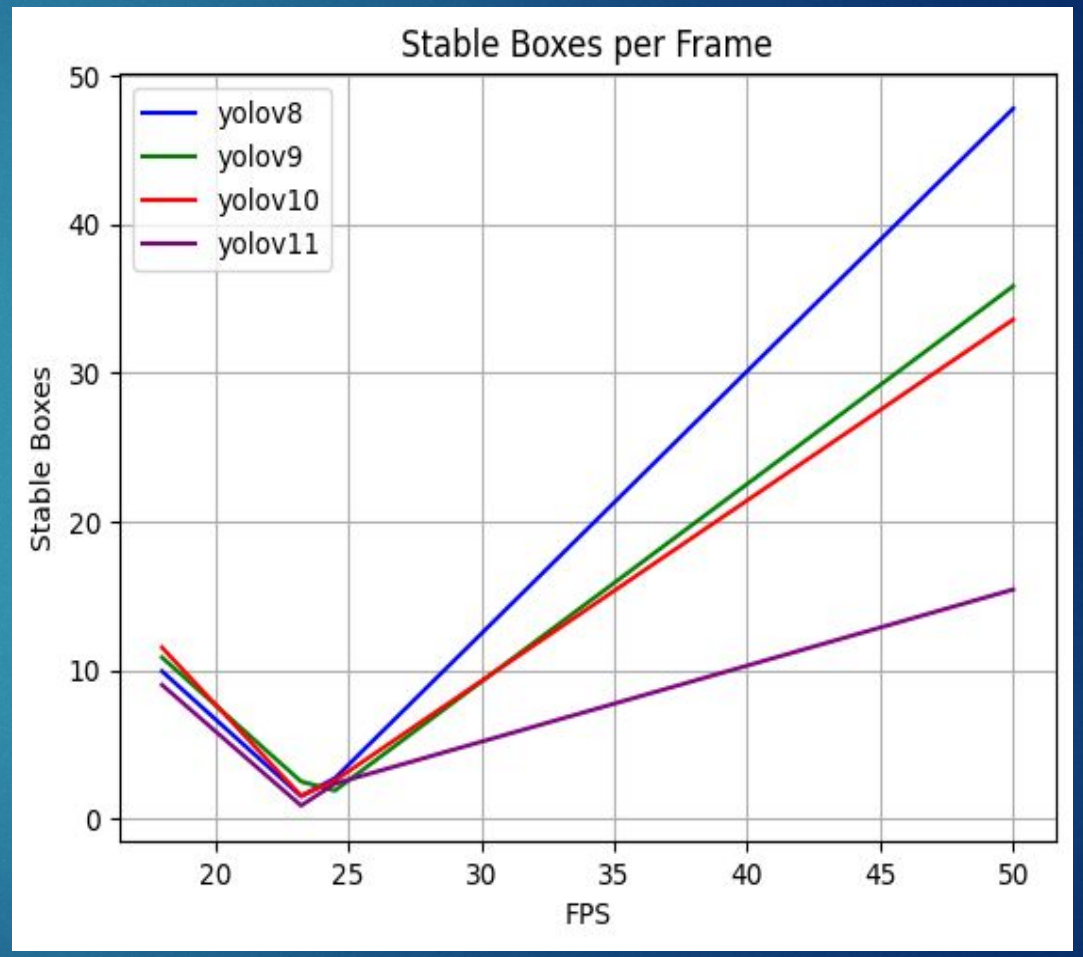

Manual Accuracy Comparison
sim_1
vid_1
vid_2
vid_3
Accuracy
Frame Sample

V11

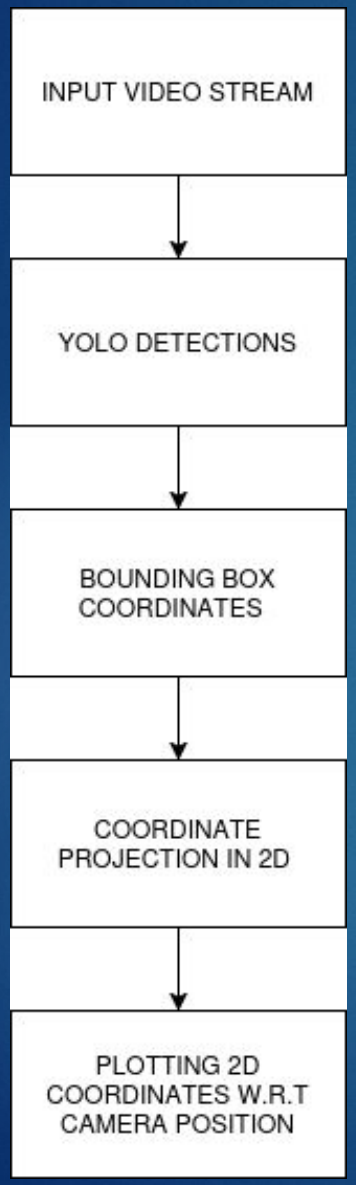

Manual Accuracy Comparison
sim_1
vid_1
vid_2
vid_3
Accuracy
Frame Sample

Impact of Video Quality?





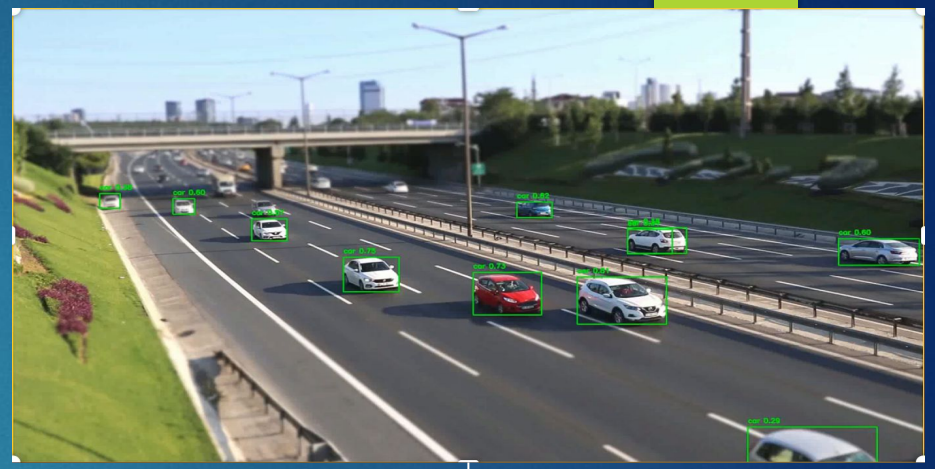
2D Point Cloud from Yolo Detections

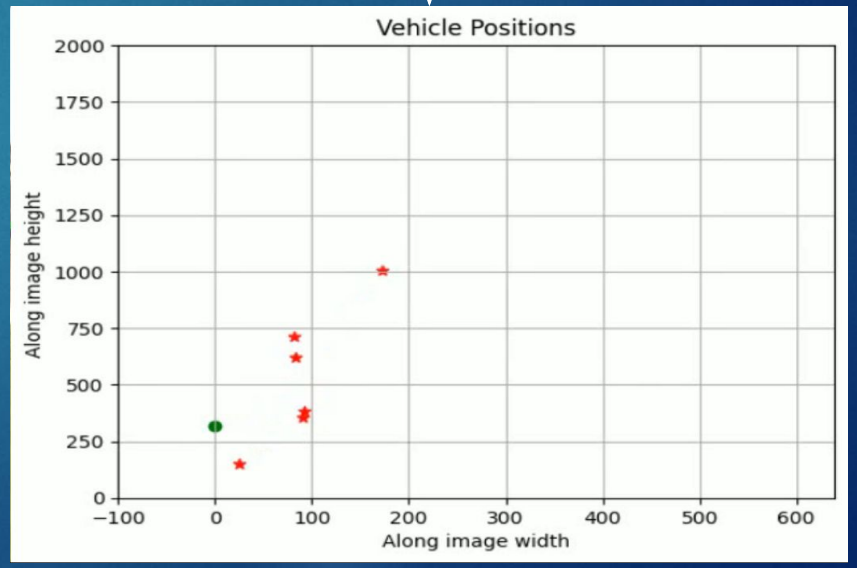


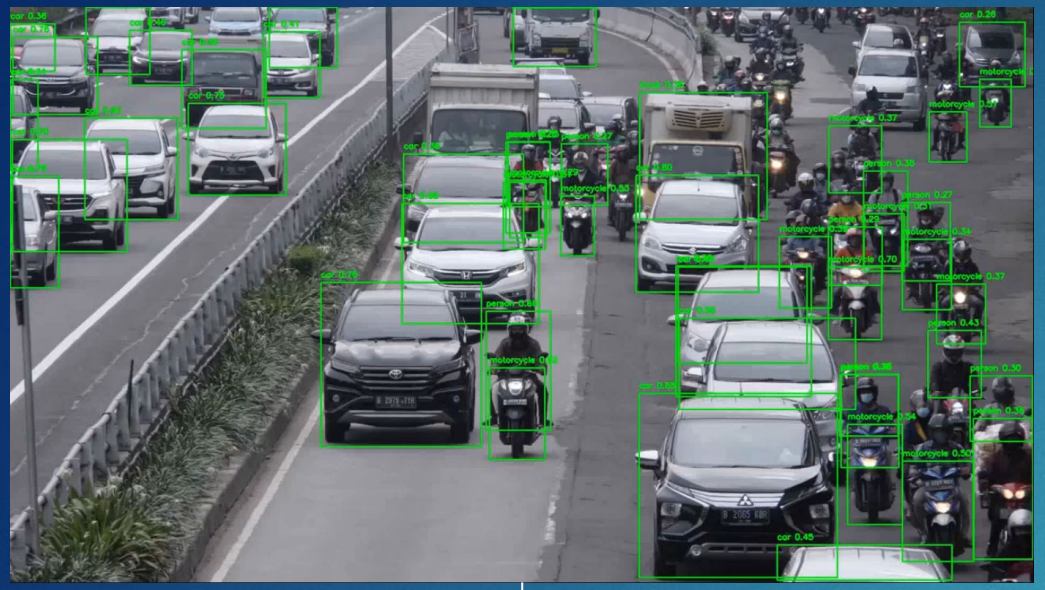
- Bounding Box Coordinates are in the form (x1,y1),(x2,y2) in a coordinate plane such that the top left corner of the image is (0,0) and the bottom right corner is (m,n)for image with mxn dimensions.

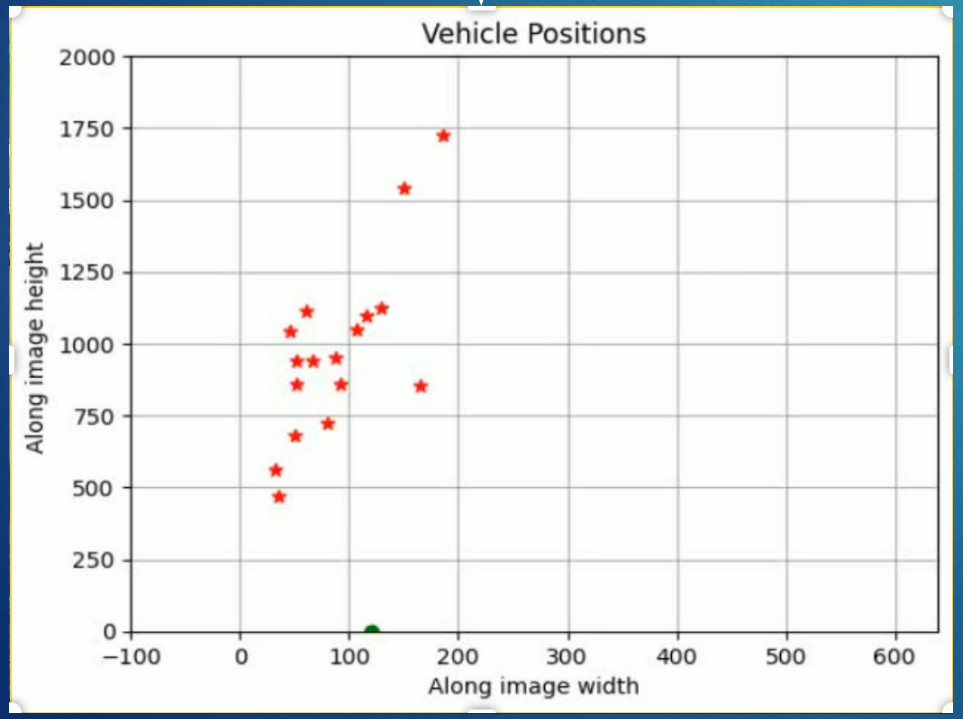
- A projection can be made based on these values

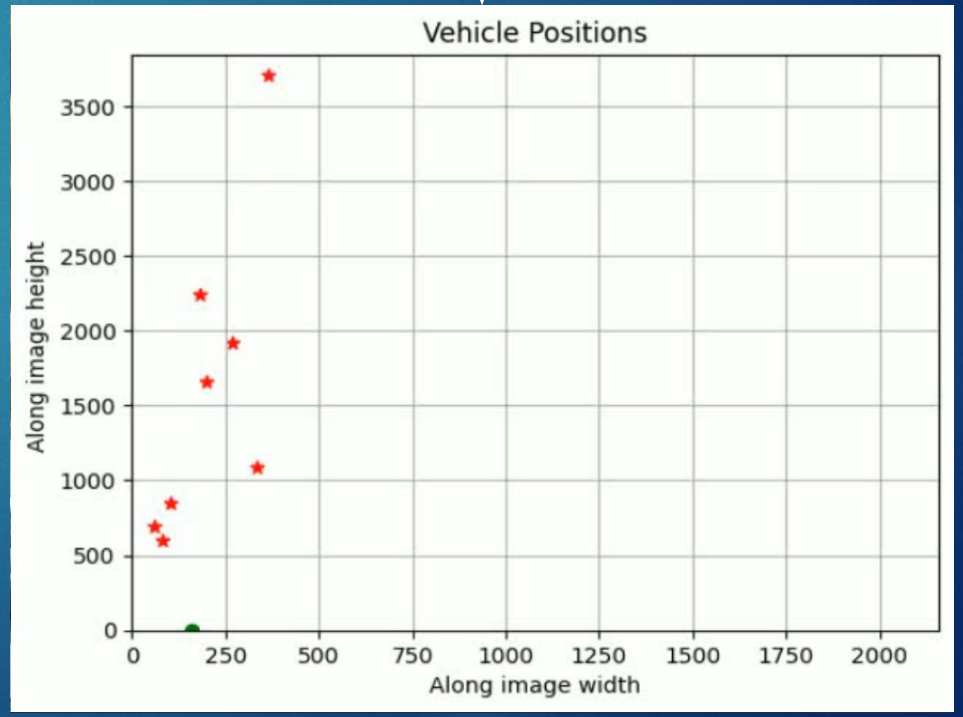
- The distance (to simulate depth) can be taken as a factor of the size of bounding box








Vehicle Positions
Along image height
Along image width
2000
1750
1500
1250
1000
750
500
250
0
−100
0
100
200
300
400
500
600


Vehicle Positions
Along image height
Along image width
3500
3000
2500
2000
1500
1000
500
0
0
250
500
750
1000
1250
1500
1750
2000

THANK YOU